1 Relation

1. The binary operation $* : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is defined as $a * b = 2a + b$. Find $(2 * 3) * 4$.

2 Matrix

1. If $A = \begin{pmatrix} 5 & a \\ b & 0 \end{pmatrix}$ is a symmetric matrix, show that $a = b$.

2. Without expanding at any stage, find the value of:

$$\begin{pmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{pmatrix}.$$

3. Use properties of determinants to solve for x:
$$\begin{pmatrix} x+a & b & c \\ c & x+b & a \\ a & b & x+c \end{pmatrix} = 0, \quad \text{and}\, x \neq 0.$$

4. Using elementary transformations, find the inverse of the matrix: $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$.

5. Using elementary transformations, find the inverse of the matrix: $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{pmatrix}$.

3 Triogonometery

1. Solve: $3\tan^{-1} x + \cot^{-1} x = \pi$.

2. If $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$, prove that $a + b + c = abc$.

3. Solve: $\sin(x - y) = \sin x \cdot \tan^{-1}\left(\frac{dy}{dx}\right)$.

4 Differentiation

1. Find the value of constant k such that the function $f(x)$ defined as: $f(x) = \begin{cases} \frac{x^2-1}{x+1}, & x \neq -1 \\ k, & x = -1 \end{cases}$ is continuous at $x = -1$.

2. Find the differential equation of the family of concentric circles $x^2+y^2 = a^2$.

3. Show that the function $f(x) = \begin{cases} x^2, & x \leq 1 \\ 2x+1, & x > 1 \end{cases}$ is continuous at $x = 1$ but not differentiable.

4. Verify Rolle's Theorem for the function $f(x) = e^{\sin x}$ on the interval $[0, \pi]$.

5. If $x = \tan(\log y)$, prove that $(1+x^2)\frac{d^2y}{dx^2} + (2x - a)\frac{dy}{dx} = 0$.

6. The population of a town grows at the rate of 10% per year. Using a differential equation, find how long it will take for the population to grow four times.

5 Geometry

1. Find the approximate change in the volume V of a cube of side x meters caused by decreasing the side by 1%.

2. Find the points on the curve $y = 4x^3 - 3x^2 + 5$ at which the equation of the tangent is parallel to the x-axis.

3. Water is dripping out from a conical funnel of semi-vertical angle θ at the uniform rate of 2 cm^2/sec through a tiny hole at the vertex. When the slant height of the water level is 4 cm, find the rate of decrease of the slant height of the water.

4. A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is maximum, find its height.

5. From a lot of 6 items containing 2 defective items, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the sample is drawn without replacement, find:

   (a) The probability distribution of X.

   (b) The mean of X.

   (c) The variance of X.

6. Draw a rough sketch of the curve and find the area of the region bounded by the curve $y = 8x$ and the line $x = 2$.

7. Sketch the graph of $y = |x+4|$. Using integration, find the area of the region bounded by the curve $y = |x+4|$ and $x = -6$ and $x = 0$.

8. Find the coefficient of correlation from the regression lines: $x - 2y + 3 = 0$ and $4x - 5y + 1 = 0$.

6 Integration

1. Evaluate: $\int \frac{1}{\sqrt{x^2+1}}\, dx$.

2. Evaluate: $\int \tan(\sqrt{x})\, dx$.

3. Evaluate: $\int \frac{\sqrt{x}}{1+x}\, dx$.

4. Evaluate: $\int \frac{x}{(1+x^2)^2}\, dx$.

7 Probability

1. If A and B are events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$, and $P(A \cap B) = \frac{1}{8}$, then find:

   (a) $P(A|B)$

   (b) $P(B|A)$

2. In a race, the probabilities of A and B winning the race are $\frac{1}{3}$ and $\frac{1}{5}$ respectively. Find the probability of neither of them winning the race.

3. A speaks truth in 60% of cases, while B speaks truth in 40% of cases. In what percentage of cases are they likely to contradict each other in stating the same fact?

4. Find the line of regression of y on x from the following table:

x	1	2	3	4	5
y	7	6	5	4	3

   Hence, estimate the value of y when $x = 6$.

5. From the given data:

Variable	Mean	Standard Deviation
x	6	4
y	8	6

and correlation coefficient $r = 0.5$. Find:

(a) Regression coefficients b_{xy} and b_{yx}.

(b) Regression line x on y.

(c) Most likely value of x when $y = 14$.

8 Functions

1. If the function $f(x) = \sqrt{2x-3}$ is invertible, find its inverse. Hence, prove that $(f \circ f^{-1})(x) = x$.

9 Vectors

1. Find λ if the scalar projection of $\mathbf{a} = \lambda\hat{\imath} + \hat{\jmath} + 4\hat{k}$ on $\mathbf{b} = 2\hat{\imath} + 6\hat{\jmath} + 3\hat{k}$ is 4 units.

2. The Cartesian equation of a line is: $2x - 3 = 3y + 1 = 5 - 6z$. Find the vector equation of a line passing through $(7, -5, 0)$ and parallel to the given line.

3. Find the equation of the plane through the intersection of the planes $\mathbf{r} \cdot (\hat{\imath} + 3\hat{\jmath} - \hat{k}) = 9$ and $\mathbf{r} \cdot (2\hat{\imath} - \hat{\jmath} + \hat{k}) = 3$ and passing through the origin.

4. If A, B, and C are three non-collinear points with position vectors $\mathbf{a}$, $\mathbf{b}$, and$\mathbf{c}$ respectively, show that the length of the perpendicular from C on AB is given by: $\frac{|\mathbf{a}\times\mathbf{b}|}{|\mathbf{a}-\mathbf{b}|}$.

5. Show that the four points A, B, C, and D with position vectors $4\hat{\imath} + 5\hat{\jmath} + \hat{k}$, $-\hat{\jmath} - \hat{k}$, $3\hat{\imath} + 9\hat{\jmath} + 4\hat{k}$, and $4(-\hat{\imath} + \hat{\jmath} + \hat{k})$ respectively, are coplanar.

6. Find the image of a point having position vector $3\hat{\imath} - 2\hat{\jmath} + \hat{k}$ in the plane $\mathbf{r} \cdot (3\hat{\imath} - \hat{\jmath} + 4\hat{k}) = 2$.

10 Calculus

1. Given the total cost function for x units of a commodity as: $C(x) = x^3 + 3x^2 - 16x + 2$. Find:

   (a) Marginal cost function.

   (b) Average cost function.

2. The average cost function associated with producing and marketing x units of an item is given by: $AC = 2x - 11 + \frac{1}{x}$. Find the range of values of the output x, for which AC is increasing.

3. A manufacturer's marginal cost function is $C'(x) = \sqrt{x} + 2$. Find the cost involved to increase production from 100 units to 300 units.

11 Linear Programming

1. A product can be manufactured at a total cost $C(x) = x^2 + 100x + 40$, where x is the number of units produced. The price at which each unit can be sold is given by $P = 200 - \frac{x}{2}$. Determine the production level x at which the profit is maximum. What is the price per unit and total profit at this level of production?

2. A manufacturing company makes two types of teaching aids A and B of Mathematics for Class X. Each type of A requires 9 labor hours for fabricating and 1 labor hour for finishing. Each type of B requires 12 labor hours for fabricating and 3 labor hours for finishing. For fabricating and finishing, the maximum labor hours available per week are 180 and 30 respectively. The company makes a profit of 80 Rupees on each piece of type A and 120 Rupees on each piece of type B. How many pieces of type A and type B should be manufactured per week to get a maximum profit? Formulate this as a Linear Programming Problem and solve it. Identify the feasible region from the rough sketch.